2 ALARM NUMBERS AND BRIEF DESCRIPTIONS

2.1 FOR Series 15*i*

2.1.1 Servo Alarm

Alarm No.	Description	Remark
SV0027	Invalid digital servo parameter setting	3.2.6
SV0361	Pulsecoder phase error (built-in)	3.2.7 (1)
SV0364	Soft phase alarm (built-in)	3.2.7 (1)
SV0365	LED error (built-in)	3.2.7 (1)
SV0366	Pulse error (built-in)	3.2.7 (1)
SV0367	Count error (built-in)	3.2.7 (1)
SV0368	Serial data error (built-in)	3.2.7 (3)
SV0369	Data transfer error (built-in)	3.2.7 (3)
SV0380	LED error (separate)	3.2.7 (2)
SV0381	Pulsecoder phase error (separate)	3.2.7 (2)
SV0382	Count error (separate)	3.2.7 (2)
SV0383	Pulse error (separate)	3.2.7 (2)
SV0384	Soft phase alarm (separate)	3.2.7 (2)
SV0385	Serial data error (separate)	3.2.7 (3)
SV0386	Data transfer error (separate)	3.2.7 (3)
SV0387	Sensor error (separate)	3.2.7 (2)
SV0421	Excessive semi-full error	3.2.8
SV0430	Servo motor overheat	3.2.5
SV0432	Converter: control power supply undervoltage	3.1.4
SV0433	Converter: DC link undervoltage	3.1.1
SV0436	Soft thermal (OVC)	3.2.3
SV0438	Inverter: motor current alarm	3.1.9
SV0439	Converter: DC link overvoltage	3.1.2
SV0440	Converter: Excessive deceleration power	3.1.3
SV0441	Current offset error	3.2.8
SV0444	Inverter: internal cooling fan stopped	3.1.5
SV0445	Soft disconnection alarm	3.2.4
SV0447	Hard disconnection alarm (separate)	3.2.4
SV0448	Feedback mismatch alarm	3.2.8
SV0449	Inverter: IPM alarm	3.1.7
SV0601	Inverter: cooling fan stopped of the radiator	3.1.6
SV0603	Inverter: IPM alarm (OH)	3.1.8

2.2 FOR Series 16*i*, 18*i*, 20*i*, 21*i*, 0*i*, AND Power Mate *i*

2.2.1 Servo Alarm

Alarm No.	Description	Remark
361	Pulsecoder phase error (built-in)	3.2.7 (1)
364	Soft phase alarm (built-in)	3.2.7 (1)
365	LED error (built-in)	3.2.7 (1)
366	Pulse error (built-in)	3.2.7 (1)
367	Count error (built-in)	3.2.7 (1)
368	Serial data error (built-in)	3.2.7 (3)
369	Data transfer error (built-in)	3.2.7 (3)
380	LED error (separate)	3.2.7 (2)
381	Pulsecoder phase error (separate)	3.2.7 (2)
382	Count error (separate)	3.2.7 (2)
383	Pulse error (separate)	3.2.7 (2)
384	Soft phase alarm (separate)	3.2.7 (2)
385	Serial data error (separate)	3.2.7 (3)
386	Data transfer error (separate)	3.2.7 (3)
387	Sensor error (separate)	3.2.7 (2)
417	Invalid parameter	3.2.6
421	Excessive semi-full error	3.2.8
430	Servomotor overheat	3.2.5
432	Converter: control undervoltage	3.1.4
433	Converter: DC link undervoltage	3.1.1
436	Soft thermal (OVC)	3.2.3
438	Inverter: motor current alarm	3.1.9
439	Converter: DC link overvoltage	3.1.2
440	Converter: excessive deceleration power	3.1.3
441	Current offset error	3.2.8
444	Inverter: internal cooling fan stopped	3.1.5
445	Soft disconnection alarm	3.2.4
447	Hard disconnection alarm (separate)	3.2.4
448	Feedback mismatch alarm	3.2.8
449	Inverter: IPM alarm	3.1.7
453	Soft disconnection alarm (α Pulsecoder)	3.2.4
601	Inverter: cooling fan stopped of the radiator	3.1.6
603	Inverter: IPM alarm (OH)	3.1.8

3 TROUBLESHOOTING AND ACTION

3.1 SERVO AMPLIFIER MODULE

The following table lists alarms related to the servo amplifier module. See this table and the CNC alarm code indicated in Chapter 2, "ALARM NUMBERS AND BRIEF DESCRIPTIONS".

Alarm	LED display	Major cause	Reference
Converter: DC link undervoltage	ON	Voltage drop at the DC link in the main circuit	3.1.1
Converter: DC link overvoltage	ON	Voltage rise at the DC link in the main circuit	3.1.2
Converter: excessive deceleration power	ON	- Too large regenerative discharge amount - Abnormal regenerative discharge circuit	3.1.3
Converter: control power supply undervoltage	ON	- Drop in external control power supply (24 V) - Connector/cable (CXA19A, CXA19B) defective - SVM failure	3.1.4
Inverter: internal cooling fan stopped	ON	- Fan not running. - Fan motor connector or cable defective - SVM failure	3.1.5
Inverter: cooling fan stopped of the radiator	ON	- Fan not running. - Fan motor connector or cable defective - SVM failure	3.1.6
Inverter: IPM alarm	ON	- Short-circuit between power lead phases or ground fault in them - Short-circuit between motor winding phases or ground fault in them - SVM failure	3.1.7
Inverter: IPM alarm (OH)	ON	- The motor is being used under a harsh condition. - The ambient temperature is high. - SVM failure	3.1.8
Inverter: motor current alarm	ON	- Short-circuit between power lead phases or ground fault in them - Short-circuit between motor winding phases or ground fault in them - Incorrect motor ID setting - SVM or motor failure	3.1.9
Inverter: FSSB communication error	ON	- Connector or cable failure - SVM or CNC failure	3.1.10

3.1.1 Converter: DC Link Undervoltage

(1) Meaning
The voltage at the DC link of the converter is low.

(2) Cause and troubleshooting
(a) A small power dip has occurred.
→ Check the power supply.
(b) Low input power supply voltage
→ Check the power supply specification.
(c) Insert the SVM face plate (control printed-circuit board) securely.
(d) Replace the SVM.

3.1.2 Converter: DC Link Overvoltage

(1) Meaning
In the main circuit, the voltage at the DC link is abnormally high.

(2) Cause and troubleshooting
(a) SVM1-4*i*, SVM1-20*i*
Use a regenerative resistor.
(b) Excessive regenerated power
Increase the acceleration/deceleration time constant. If this alarm occurs less frequently, the regeneration capacity is not sufficient. Set the acceleration/deceleration time constant to such a level that does not cause any alarm.
(c) Insert the SVM face plate (control printed-circuit board) securely.
(d) Replace the SVM.

3.1.3 Converter: Excessive Deceleration Power

(1) Meaning
The converter deceleration power is too large.

(2) Cause and troubleshooting
For SVM1-4*i* and SVM1-20*i*

- When a separate regenerative resistor is not used
  (a) Connect CXA20 by using a dummy connector.
  (b) Insert the SVM face plate (control printed-circuit board) securely.
  (c) Replace the SVM.
- When a separate regenerative resistor is used
  (a) Check the resistance at both ends of connector CXA20 on the regenerative resistor side to confirm that the resistance is 0 Ω.
  (b) The average regenerative power may be high. Decrease the frequency of acceleration/deceleration, or review the resistor specification.
  (c) Insert the SVM face plate (control printed-circuit board) securely.
  (d) Replace the SVM.

For SVM1-40*i* and SVM1-80*i*

- When the built-in regenerative resistor is used
  - (a) Connect CXA20 and CZ6 by using a dummy connector.
  - (b) The average regenerative power may be high. Decrease the frequency of acceleration/deceleration, or review the resistor specification.
  - (c) Insert the SVM face plate (control printed-circuit board) securely.
  - (d) Replace the SVM.

- When a separate regenerative resistor is used
  - (a) Check the resistance at both ends of connector CXA20 on the regenerative resistor side to confirm that the resistance is 0 Ω.
  - (b) The average regenerative power may be high. Decrease the frequency of acceleration/deceleration, or review the resistor specification.
  - (c) Insert the SVM face plate (control printed-circuit board) securely.
  - (d) Replace the SVM.

3.1.4 Converter: Control Power Supply Undervoltage

(1) Meaning
The external control power supply (24 VDC) voltage is low.

(2) Cause and troubleshooting
- (a) Check the voltage level of the external power supply (24 VDC). (Normal voltage: 21.6 V or higher)
- (b) Check the connector and cable (CXA19A, CXA19B).
- (c) Replace the SVM.

3.1.5 Inverter: Internal Cooling Fan Stopped

(1) Meaning
Inverter: internal cooling fan stopped

(2) Cause and troubleshooting
- (a) Check whether there is any foreign material in the fan.
- (b) Be sure to push the faceplate (control printed-circuit board) as far as it will go.
- (c) Check that the fan connector is attached correctly.
- (d) Replace the fan.
- (e) Replace the SVM.

3.1.6 Inverter: Cooling Fan Stopped of the Radiator

(1) Meaning
Inverter: cooling fan stopped of the radiator

(2) Cause and troubleshooting
- (a) Check whether there is any foreign material in the fan.
- (b) Be sure to push the faceplate (control printed-circuit board) as far as it will go.
- (c) Check that the fan connector is attached correctly.
- (d) Replace the fan.
- (e) Replace the SVM.

3.1.7 Inverter: IPM Alarm

(1) Meaning
Inverter: IPM alarm

(2) Cause and troubleshooting
- (a) Be sure to push the faceplate (control printed-circuit board) as far as it will go.
- (b) Disconnect the motor power leads from the SVM, and release the SVM from an emergency stop condition.
  - <1> If no IPM alarm condition has occurred
    → Go to (c).
  - <2> If an IPM alarm condition has occurred
    → Replace the SVM.
- (c) Disconnect the motor power leads from the SVM, and check the insulation between PE and the motor power lead U, V, or W.
  - <1> If the insulation is deteriorated
    → Go to (d).
  - <2> If the insulation is normal
    → Replace the SVM.
- (d) Disconnect the motor from its power leads, and check whether the insulation of the motor or power leads is deteriorated.
  - <1> If the insulation of the motor is deteriorated
    → Replace the motor.
  - <2> If the insulation of any power lead is deteriorated
    → Replace the power lead.

3.1.8 Inverter: IPM Alarm (OH)

(1) Meaning
Inverter: IPM alarm (OH)

(2) Cause and troubleshooting
- (a) Be sure to push the faceplate (control printed-circuit board) as far as it will go.
- (b) Check that the heat sink cooling fan is running.
- (c) Check that the motor is being used at or below its continuous rating.
- (d) Check that the cooling capacity of the cabinet is sufficient (inspect the fans and filters).
- (e) Check that the ambient temperature is not too high.
- (f) Replace the SVM.

3.1.9 Inverter: Motor Current Alarm

(1) Meaning
Inverter: DC link current alarm

(2) Cause and troubleshooting

(a) Checking the servo parameters
Referring to "FANUC AC SERVO MOTOR αis/αi/βis series Parameter Manual (B-65270EN)," check whether the following parameters have default values.

Series 15*i*	No.1809	No.1852	No.1853
Series 16*i*, 18*i*, 20*i*, 21*i*, 0*i* Power Mate *i*	No.2004	No.2040	No.2041

Alternatively, if an abnormal motor current alarm condition occurs only on rapid acceleration/deceleration, it is likely that the motor is being used under too harsh a condition. Increase the acceleration/deceleration time constant, and see what will occur.

(b) Be sure to push the faceplate (control printed-circuit board) as far as it will go.

(c) Disconnect the motor power leads from the SVM, and release the SVM from an emergency stop condition.
<1> If no abnormal motor current occurs
→ Go to (d).
<2> If an abnormal motor current occurs
→ Replace the SVM.

(d) Disconnect the motor power leads from the SVM, and check the insulation between PE and the motor power lead U, V, or W.
<1> If the insulation is deteriorated
→ Go to (e).
<2> If the insulation is normal
→ Replace the SVM.

(e) Disconnect the motor from its power leads, and check whether the insulation of the motor or power leads is deteriorated.
<1> If the insulation of the motor is deteriorated
→ Replace the motor.
<2> If the insulation of any power lead is deteriorated
→ Replace the power lead.

3.1.10 Inverter: FSSB Communication Error

(1) Meaning
Inverter: FSSB communication error

(2) Cause and troubleshooting
(a) Replace the optical cable (COP10A) of the SVM that is nearest to the CNC among the amplifiers on which the ALM LED is lit (in Fig. 3.1.10, the cable between UNIT2 and UNIT3).
(b) Replace the SVM that is the second nearest to the CNC among the amplifiers on which the ALM LED is lit (in Fig. 3.1.10, UNIT3).
(c) Replace the SVM that is nearest to the CNC among the amplifiers on which the ALM LED is lit (in Fig. 3.1.10, UNIT2).
(d) Replace the servo card in the CNC.

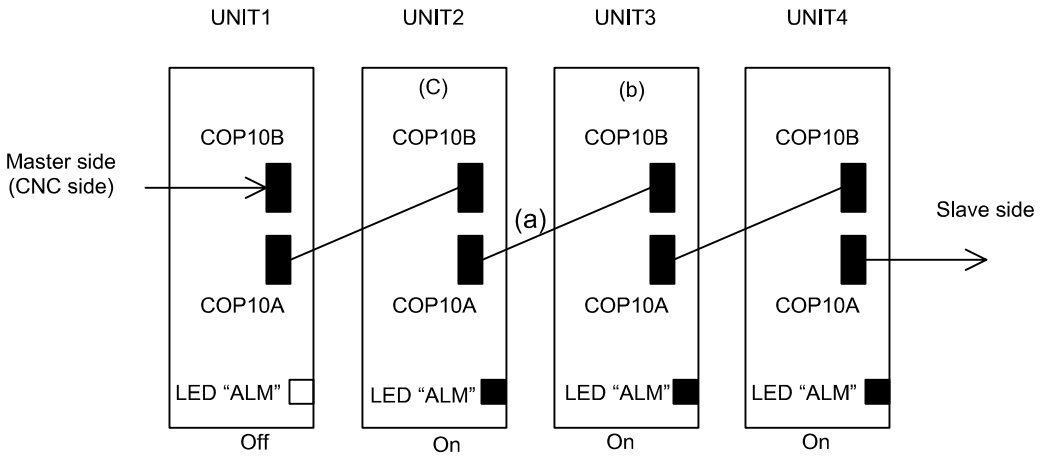

Fig. 3.1.10

3.2 SERVO SOFTWARE

If a servo alarm is issued, an alarm message is output, and details of the alarm are also displayed on the servo adjustment screen or the diagnosis screen. Using the alarm identification table given in this section, determine the alarm, and take a proper action.

3.2.1 Servo Adjustment Screen

The following procedure can be used to display the servo adjustment screen.
(The DPL/MDI of the Power Mate has no servo adjustment screen.)

- **Series 15*i***

SYSTEM → [CHAPTER] → [SERVO] → [▷] → [SERVO ALARM]

- **Series 16*i*, 18*i*, 20*i*, 21*i*, 0*i***

SYSTEM → [SYSTEM] → [▷] → [SV-PRM] → [SV-TUN]

If the servo setting screen does not appear, specify the following parameter, then switch the CNC off and on again.

	#7	#6	#5	#4	#3	#2	#1	#0
3111								SVS

SVS (#0)=1 (to display the servo setting screen)

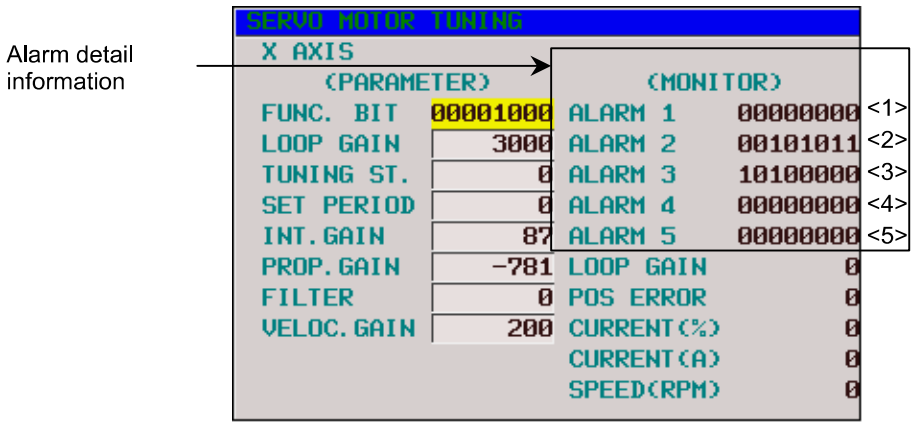

Fig. 3.2.1(a) Servo adjustment screen

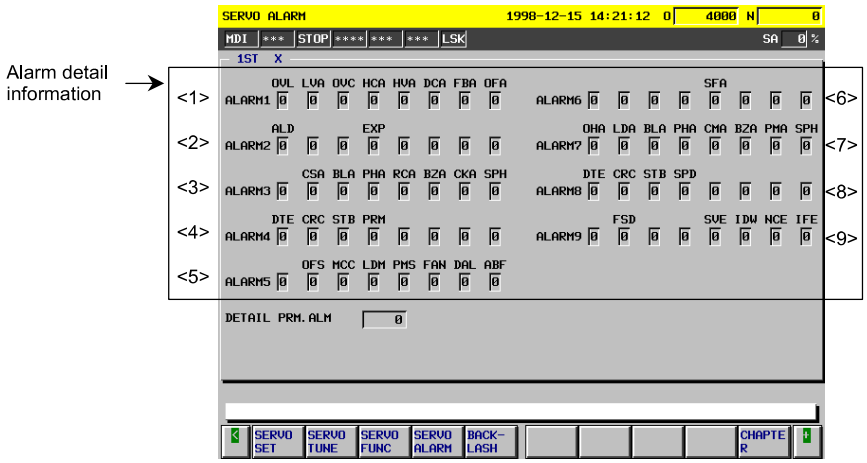

Fig. 3.2.1(b) Series 15*i* servo alarm screen

The table below indicates the names of the alarm bits.

Table 3.2.1 List of alarm bit names

	#7	#6	#5	#4	#3	#2	#1	#0
<1> Alarm 1	OVL	LVA	OVC	HCA	HVA	DCA	FBA	OFA
<2> Alarm 2	ALD			EXP				
<3> Alarm 3		CSA	BLA	PHA	RCA	BZA	CKA	SPH
<4> Alarm 4	DTE	CRC	STB	PRM				
<5> Alarm 5		OFS	MCC	LDM	PMS	FAN	DAL	ABF
<6> Alarm 6					SFA			
<7> Alarm 7	OHA	LDA	BLA	PHA	CMA	BZA	PMA	SPH
<8> Alarm 8	DTE	CRC	STB	SPD				
<9> Alarm 9		FSD			SVE	IDW	NCE	IFE

NOTE
The empty fields do not represent alarm codes.

3.2.2 Diagnosis Screen

The alarm items of the servo adjustment screen correspond to the diagnosis screen numbers indicated in the table below.

Table 3.2.2 Correspondence between the servo adjustment screen and diagnosis screen

Alarm No.	Series 15*i*	Series 16*i*, 18*i*, 21*i*, 0*i*
<1> Alarm 1	No 3014 + 20(X-1)	No 200
<2> Alarm 2	3015 + 20(X-1)	201
<3> Alarm 3	3016 + 20(X-1)	202
<4> Alarm 4	3017 + 20(X-1)	203
<5> Alarm 5	————	204
<6> Alarm 6	————	———
<7> Alarm 7	————	205
<8> Alarm 8	————	206
<9> Alarm 9	————	———

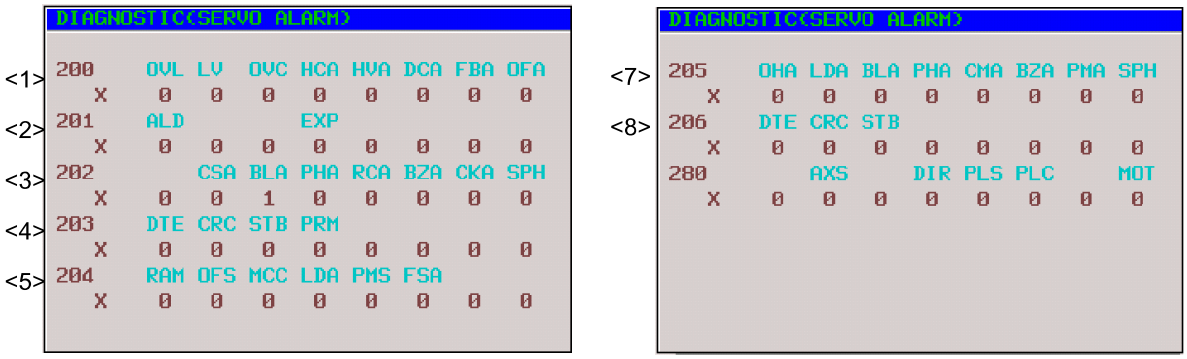

Fig. 3.2.2 Diagnosis screen

3.2.3 Overload Alarm (Soft Thermal, OVC)

(Alarm identification method)

	#7	#6	#5	#4	#3	#2	#1	#0
<1> Alarm 1	OVL	LVA	OVC	HCA	HVA	DCA	FBA	OFA

(Action)

(1) Make sure that the motor is not vibrating.
⇒ If a motor vibrates, the current flowing in it becomes more than necessary, resulting in an alarm.

(2) Make sure that the power lead to the motor is connected correctly.
⇒ If the connection is incorrect, an abnormal current flows in the motor, resulting in an alarm.

(3) Make sure that the following parameters are set correctly.
⇒ An overload alarm is issued based on the result of calculation of these parameters. Be sure to set them to the standard values. For details of the standard values, refer to the FANUC AC SERVO MOTOR αis/αi/βis series Parameter Manual (B-65270EN).

No. 1877 (FS15*i*) No. 2062 (FS16*i*)	Overload protection coefficient (OVC1)
No. 1878 (FS15*i*) No. 2063 (FS16*i*)	Overload protection coefficient (OVC2)
No. 1893 (FS15*i*) No. 2065 (FS16*i*)	Overload protection coefficient (OVCLMT)
No. 1785 (FS15*i*) No. 2162 (FS16*i*)	Overload protection coefficient (OVC21)
No. 1786 (FS15*i*) No. 2163 (FS16*i*)	Overload protection coefficient (OVC22)
No. 1787 (FS15*i*) No. 2164 (FS16*i*)	Overload protection coefficient (OVCLMT2)

(4) Attach the check board to connector JX5 to measure the waveform of the actual current (IR and IS) of the servo amplifier module. (This check pin board differs from that for the α series.) Run the motor and measure its actual currents (IR and IS). Then compare the measurement results with the overload duty curve shown in "FANUC AC SERVO MOTOR βis series Descriptions (B-65302EN)," and see whether the machine load is too heavy compared with the motor capacity. If the actual current is high on acceleration/deceleration, it is likely that the time constant is too small.

3.2.4 Feedback Disconnected Alarm

(Alarm identification method)

	#7	#6	#5	#4	#3	#2	#1	#0
<1> Alarm 1	OVL	LVA	OVC	HCA	HVA	DCA	FBA	OFA
<2> Alarm 2	ALD			EXP				
<6> Alarm 6					SFA			

FBA	ALD	EXP	SFA	Alarm description	Action
1	1	1	0	Hard disconnection (separate phase A/B)	1
1	0	0	0	Soft disconnection (closed loop)	2
1	0	0	1	Soft disconnection (αi Pulsecoder)	3

(Action)

Action 1: This alarm is issued when a separate phase A/B scale is used. Check if the phase A/B detector is connected correctly.

Action 2: This alarm is issued when the position feedback pulse variation is small relative to the velocity feedback pulse variation. This means that this alarm is not issued when a semi-full is used. Check if the separate detector outputs position feedback pulses correctly. If position feedback pulses are output correctly, it is considered that the motor alone is rotating in the reverse direction at the start of machine operation because of a large backlash between the motor position and scale position.

	#7	#6	#5	#4	#3	#2	#1	#0
No. 1808 (FS15*i*)							TGAL	
No. 2003 (FS16*i*)								

TGAL (#1) 1: Uses the parameter for the soft disconnection alarm detection level.

No. 1892 (FS15*i*)	Soft disconnection alarm level
No. 2064 (FS16*i*)	

Standard setting 4: Alarm issued for a 1/8 rotation of the motor. Increase this value.

Action 3: This alarm is issued when synchronization between the absolute position data sent from the built-in Pulsecoder and phase data is lost. Turn off the power to the CNC, then detach the Pulsecoder cable then attach it again. If this alarm is still issued, replace the Pulsecoder.

3.2.5 Overheat Alarm

(Alarm identification method)

	#7	#6	#5	#4	#3	#2	#1	#0
<1> Alarm 1	OVL	LVA	OVC	HCA	HVA	DCA	FBA	OFA
<2> Alarm 2	ALD			EXP				

OVL	ALD	EXP	Alarm description	Action
1	1	0	Motor overheat	1
1	0	0	Amplifier overheat	1

(Action)

Action 1: If this alarm is issued after a long-time of continuous operation, it is considered that the motor and amplifier are overheated. Stop operation for a while, then make a check. If this alarm is still issued after the power is off for about 10 minutes then is turned on again, the thermostat is considered to be faulty. If this alarm is issued intermittently, increase the time constant or increase stop time in the program to suppress the rise in temperature.

3.2.6 Invalid Servo Parameter Setting Alarm

The invalid servo parameter setting alarm is issued when a setting out of the specifiable range is specified, or an overflow has occurred in an internal calculation. When an invalid parameter is detected on the servo side, alarm 4 #4 (PRM) = 1 results.

(Alarm identification method)

	#7	#6	#5	#4	#3	#2	#1	#0
<4> Alarm 4	DTE	CRC	STB	PRM				

For details and action required when the invalid servo parameter setting alarm is issued on the servo side, refer to the FANUC AC SERVO MOTOR αis/αi/βis series Parameter Manual (B-65270EN).

(Reference information)

Method of checking details of an invalid parameter detected on the servo side

(For Series 15*i*)

A number is indicated in the item "Details of invalid parameter" on the servo alarm screen (Fig. 3.2.1(b)).

(For Series 16*i*, 18*i*, 21*i*, 0*i*, and Power Mate *i*)

A number is indicated in No. 352 of the diagnosis screen.

3.2.7 Alarms Related to Pulsecoder and Separate Serial Detector

(Bits for alarm identification)

	#7	#6	#5	#4	#3	#2	#1	#0
<1> Alarm 1	OVL	LVA	OVC	HCA	HVA	DCA	FBA	OFA
<2> Alarm 2	ALD			EXP				
<3> Alarm 3		CSA	BLA	PHA	RCA	BZA	CKA	SPH
<4> Alarm 4	DTE	CRC	STB	PRM				
<5> Alarm 5		OFS	MCC	LDM	PMS	FAN	DAL	ABF
<6> Alarm 6					SFA			
<7> Alarm 7	OHA	LDA	BLA	PHA	CMA	BZA	PMA	SPH
<8> Alarm 8	DTE	CRC	STB	SPD				
<9> Alarm 9		FSD			SVE	IDW	NCE	IFE

(1) For a built-in Pulsecoder
An alarm is determined from the bits of alarms 1, 2, 3, and 5. The table below indicates the meaning of each bit.

Alarm 3							Alarm 5		1	Alarm 2		Alarm description	Action
CSA	BLA	PHA	RCA	BZA	CKA	SPH	LDM	PMA	FBA	ALD	EXP		
						1						Soft phase alarm	2
				1								Zero battery voltage	1
			1						1	1	0	Count error alarm	2
		1										Phase alarm	2
	1											Battery voltage decrease (Caution)	1
								1				Pulse error alarm	
							1					LED error alarm	

⚠ CAUTION
An alarm for which no action number is given is considered to be caused by a Pulsecoder failure. Replace the Pulsecoder.

(2) For a separate serial detector
An alarm is determined from the bits of alarm 7. The table below indicates the meaning of each bit.

Alarm 7								Alarm description	Action
OHA	LDA	BLA	PHA	CMA	BZA	PMA	SPH		
							1	Soft phase alarm	2
						1		Pulse error alarm	
					1			Zero battery voltage	1
				1				Count error alarm	2
			1					Phase alarm	2
		1						Battery voltage decrease (Caution)	1
	1							LED error alarm	
1								Separate detector alarm	**3**

> ⚠ **CAUTION**
> An alarm for which no action number is given is considered to be caused by a detector failure. Replace the detector.

(Action)

Action 1: Battery-related alarms
Check if a battery is connected. When the power is turned on for the first time after a battery is connected, the zero battery voltage alarm is issued. In such a case, turn off the power, then turn on the power again. If the alarm is still issued, check the battery voltage. If the battery voltage decrease alarm is issued, check the voltage, and replace the battery as required.

Action 2: Alarms that may be issued for noise
If an alarm is issued intermittently or after emergency stop cancellation, noise is probably the cause. So, provide noise protection. If the same alarm is still issued after noise protection is provided, replace the detector.

Action 3: Alarm condition detected by the separate detector
If the separate detector detects an alarm condition, contact the manufacturer of the detector for information on troubleshooting.

(3) Alarms related to serial communication
An alarm is determined from the bits of alarms 4 and 8.

Alarm 4			Alarm 8			Alarm description
DTE	CRC	STB	DTE	CRC	STB	
1						Serial Pulsecoder communication alarm
	1					
		1				
			1			Separate serial Pulsecoder communication alarm
				1		
					1	

Action: Serial communication is not performed correctly. Check if the cable is connected correctly and is not broken. If CRC or STB is issued, noise may be the cause. So, provide noise protection. If CRC or STB is always issued after the power is turned on, the Pulsecoder or amplifier control board or the pulse module may be faulty.

3.2.8 Other Alarms

(Alarm identification method)

	#7	#6	#5	#4	#3	#2	#1	#0
<5> Alarm 5		OFS	MCC	LDM	PMS	FAN	DAL	ABF

OFS	DAL	ABF	Alarm description	Action
		1	Feedback mismatch alarm	1
	1		Excessive semi-full error alarm	2
1			Current offset error alarm	3

(Action)

Action 1: This alarm is issued when the move direction of the position detector is opposite to the move direction of the speed detector. Check the rotation direction of the separate detector. If the rotation direction of the separate detector is opposite to the rotation direction of the motor, take the following action:

For a phase A/B detector: Reverse the connections of A and $\overline{A}$.

For a serial detector: Reverse the setting of the signal direction of the separate detector.

In the Series 90B0/G(07) and subsequent editions, the following settings enable signal directions in the A/B phase detector to be inverted.

	#7	#6	#5	#4	#3	#2	#1	#0
No. 1960 (FS15*i*)								RVRSE
No. 2018 (FS16*i*)								

RVRSE (#0) Reverses the signal direction of the separate detector.
0: Does not reverse the signal direction of the separate detector.
1: Reverses the signal direction of the separate detector.

If a large distortion exists between the motor and separate detector, this alarm may be issued in the case of abrupt acceleration/deceleration. In such a case, modify the detection level.

	#7	#6	#5	#4	#3	#2	#1	#0
No. 1741 (FS15*i*)							RNLV	
No. 2201 (FS16*i*)								

RNLV (#1) Modifies the feedback mismatch alarm detection level.
1: Detected with 1000 min^{-1} or more
0: Detected with 600 min^{-1} or more

Action 2: This alarm is issued when the difference between the motor position and separate detector position exceeds the excessive semi-full error level. Check if the conversion efficient for dual position feedback is set correctly. If the conversion efficient is set correctly, increase the alarm level. If this alarm is still issued after the level is modified, check the connection direction of the scale.

No. 1971 (FS15*i*)	**Dual position feedback conversion coefficient (numerator)**
No. 2078 (FS16*i*)	

No. 1972 (FS15*i*)	**Dual position feedback conversion coefficient (denominator)**
No. 2079 (FS16*i*)	

$$\text{Conversion coefficient} = \frac{\left(\begin{array}{l}\text{Number of feedback pulses per motor}\\ \text{revolution (detection unit)}\end{array}\right)}{1{,}000{,}000}$$

No. 1729 (FS15*i*)	**Dual position feedback semi-full error level**
No. 2118 (FS16*i*)	

[Setting] Detection unit. When 0 is set, no detection is made.

Action 3: The current offset value of the current detector (equivalent to the current value in the emergency stop state) is abnormally high. If this alarm is still issued after the power is turned off then back on, the current detector is faulty. For the α*i* series, replace the amplifier.